

HR_EMPLOYEE ATTRITION ANALYSIS REPORT

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Programme: Cirvee Tech Academy (Final Project)

Dataset: 1470 Employees

Tools: SQL and Excel

EXECUTIVE SUMMARY

1470	\$6.5K	16%	84%	\$4.8K
Total Employee	Avg. Monthly Income (Retention)	Attrition Rate	Retention Rate	Avg. Monthly Income (Attrition)

This report presents a comprehensive analysis of employee attrition using the HR Employee Attrition dataset comprising 1,470 employees. Using SQL for data querying and Excel for visualization, the analysis identifies key factors driving employee turnover. The organization recorded a 16% attrition rate with 237 employees leaving and 1,233 retained. Average monthly income for attrited employees was \$4.8K compared to \$6.5K for retained employees. Key findings reveal that early tenure employees, R&D department staff, overtime workers and low-income earners are most at risk of leaving. Recommendations focus on compensation review, career development and employee engagement strategies to reduce turnover.

PROBLEM STATEMENT

Employee attrition remains a critical challenge for organizations, leading to increased recruitment costs, loss of institutional knowledge and reduced productivity. This analysis seeks to **identify the key factors that contribute to employee attrition** and provide actionable recommendations on how organizations can reduce turnover and improve retention through data-driven decision making.

DATA OVERVIEW

The HR Employee Attrition dataset is a single CSV file containing employee information across 1,470 records and 35 variables. The dataset captures a wide range of employee attributes including demographic information, compensation metrics, job satisfaction, performance ratings, tenure and attrition status. Each row represents a unique employee identified by a unique Employee ID. The data was sourced from Kaggle website (Online) and was used to analyze the key factors influencing employee turnover and retention within an organization.

Data Dictionary

Variable	Type	Description	Used in Analysis
Employee_ID	INT	Unique identifier assigned to each employee	No
Age	INT	Age of the Employee in years	Yes
Gender	Varchar	Biological sex of the employee (Male and Female)	Yes
Marital Status	Varchar	Marital status of the employee (Married, Single or Divorced)	Yes
Education	INT	Education level of the employee on a scale of 1-5	Yes
Education Field	Varchar	Field of study of the employee (e.g., Life Sciences, Medical, Marketing)	No
Num. Of Companies Worked	INT	Total number of companies the employee has worked at prior to current organization	No
Over 18	Varchar	Indicates whether the employee is above 18 years of age (Yes or No)	No
Total Working Year	INT	Total number of years the employee has been working professionally	No
Training Time Last Year	INT	Number of training sessions attended by the employee in the previous year	No
Year At Company	INT	Number of years the employee has been with the current organization	No
Years In Current Role		Number of years the employee has been in their current job role	No

Year Since Last Promotion		Promotion Number of years since the employee last received a promotion	Yes
Year With Current Manager		Number of years the employee has worked under their current manager	No
Attrition		Indicates whether the employee left the organization (Yes) or stayed (No)	Yes
Business Travel		Frequency of business travel (Non-Travel, Travel Rarely or Travel Frequently)	No
Daily Rate		Daily compensation rate of the employee in dollars	No
Department		Department the employee belongs to (Sales, R&D or Human Resources)	Yes
Distance		Distance from the employee's home to the workplace (Near, Far or Very Far)	Yes
Employee Count		Total count of employees (constant value used for aggregation purposes)	No
Environment Satisfaction		Employee's satisfaction with the work environment on a scale of 1-4 (Low, Medium, High, Very High)	No
Hourly Rate		Hourly compensation rate of the employee	No
Job Involvement		Level of employee involvement in their job on a scale of 1-4 (Low, Medium, High, Very High)	No
Job Level		Seniority level of the employee within the organization on a scale of 1-5	Yes
Job Role		Specific job title or role of the employee (e.g.	Yes

		Sales Executive, Research Scientist)	
Job Satisfaction		Employee's satisfaction with their job on a scale of 1-4	Yes
Monthly Income		Total monthly salary of the employee in dollars	Yes
Monthly Rate		Monthly compensation rate used for payroll calculations	No
Over Time		Indicates whether the employee works overtime (Yes or No)	Yes
Percent Salary Hike		Percentage increase in salary received by the employee during last appraisal	No
Performance Rate		Performance rating of the employee (1-4)	Yes
Relationship Satisfaction		Employee's satisfaction with workplace relationships on a scale of 1-4	Yes
Standard Hour		Standard number of working hours per day (constant value for all employees)	No
Stock Option Level		Level of stock options granted to the employee on a scale of 0-3	No
Work life Balance		Employee's rating of their work-life balance on a scale of 1-4 (Bad, Good, Better, Best)	Yes

Data Quality Issues & Decisions

Issue	Volume	Decision
Missing Value	0%	No decision
Inconsistent format	All Text Column	Standardize all text value

METHODOLOGY

A five-stage pipeline using SQL and Excel was adopted to ensures reproducibility and separation of raw from analytical data.

Stage	Tool	Output	Key Decision
Ingestion	MYSQL	1470	Load CSV file into SQL table created in database for analysis
Exploration	MYSQL	Null count and inconstistence format	0% null found and many inconsistencies format was found like text value not standardize
Cleaning & Transformation	MYSQL	HR_Employee clean (1470 rows) + 1 new columns added	Standardize text value and replace nulls. Extracted Age Group from Age Column
Analysis	MYSQL	12 Descriptive questions answered	All Analysis was done in SQL
Visualization	Excel	Dashboard	Display insights through charts and graphs

KEY FINDINGS

1. Overall Attrition Overview

Out of 1,470 employees, 237 left while 1,233 were retained. Although the 16% attrition rate appears relatively low, the 237 exits still represent a meaningful group worth analyzing to understand potential retention issues.

Implication: Proactive retention strategies should be implemented before turnover escalates and recruitment costs increase.

2. Attrition by Gender vs Age Group

Employees aged 28-37 years recorded the highest attrition with 389 males and 249 females in that group. Male employees are more likely to leave than females, often seeking better opportunities or faster career growth.

Implication: Targeted career development programs should be introduced for employees aged 28-37 to improve retention.

3. Employee Tenure (Early Attrition)

Employees with 0-2 years tenure recorded the highest early exits at 102, followed by 3-5 years (60), 6-10 years (55) and 10+ years (20).

Implication: Onboarding programs and early engagement strategies need urgent improvement to retain new employees within their first two years.

4. Attrition by Education Level vs Monthly Income

Bachelor degree holders recorded the highest attrition at 99 employees with an average monthly income of \$4.8K, lower than Master's holders (\$5.3K) and Doctorate holders (\$5.9K).

Implication: Reviewing compensation structures for Bachelor degree holders could significantly reduce attrition in this group.

5. Attrition by Job Role vs Distance

Sales Executives recorded the highest attrition with 57 exits, particularly those living far (21) and very far (16) from the workplace.

Implication: Introducing remote work options or transport allowances for Sales Executives could reduce distance-related turnover.

6. Attrition by Job Satisfaction

Employees with Fair job satisfaction recorded the highest attrition at 47%, followed by Good (31%) and Excellent (22%).

Implication: Organizations should conduct regular satisfaction surveys and address concerns early to prevent dissatisfied employees from leaving.

7. Attrition by Department

The Research & Development department recorded the highest attrition with 133 employees leaving, making it the most affected department due to lack of challenging projects and insufficient compensation.

Implication: R&D leadership should focus on project enrichment and competitive compensation to retain top talent.

8. Attrition by Promotion Tenure

Employees promoted less than 1 year ago recorded the highest attrition at 159, followed by 1-5 years (43), 6-10 years (27) and 11-15 years (8).

Implication: Organizations should provide adequate support and mentorship for recently promoted employees to ease their transition.

9. Attrition by Performance Rating

Employees rated Excellent recorded the highest attrition at 84% compared to Outstanding performers at 16%, suggesting strong contributors feel undervalued or lack growth opportunities.

Implication: Retaining high performers requires recognition programs and competitive compensation to prevent talent drain.

10. Attrition by Overtime

Employees who work overtime recorded the highest attrition at 53.59% compared to 46.41% for non-overtime workers, indicating excessive overtime leads to burnout.

Implication: Organizations should enforce reasonable working hours and workload management policies to reduce burnout-related attrition.

11. Attrition by Marital Status vs Monthly Income

Single employees recorded the highest attrition count of 120 with a low average monthly income of \$4.4K.

Implication: Targeted compensation and retention strategies for single employees could significantly reduce turnover in this demographic.

12. Attrition by Work-Life Balance

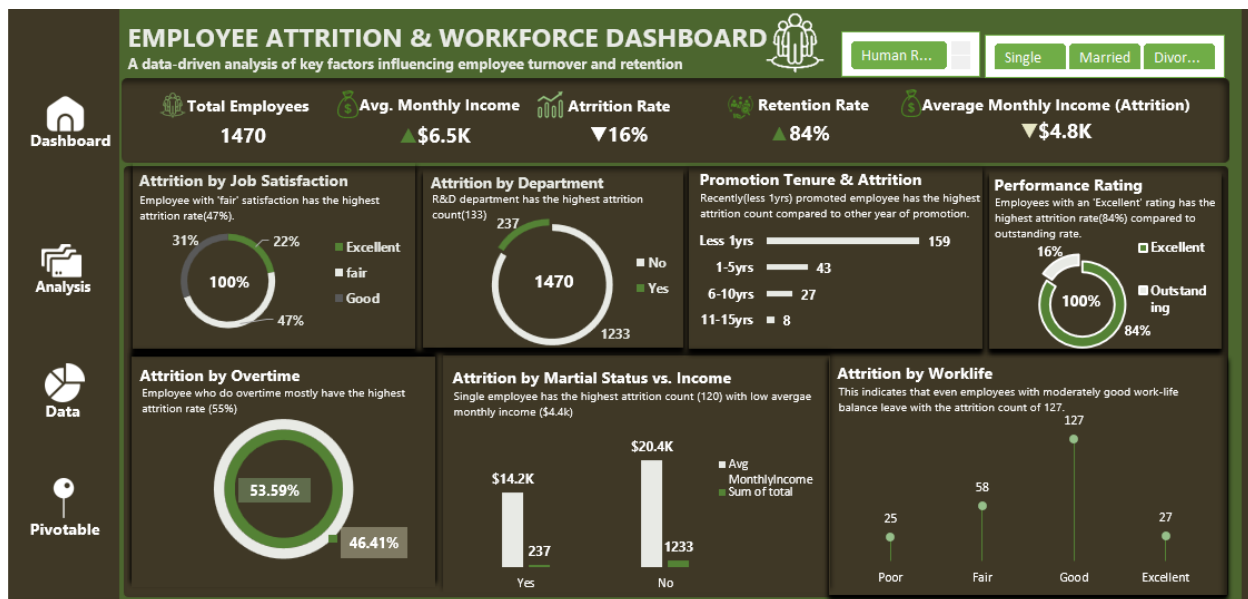
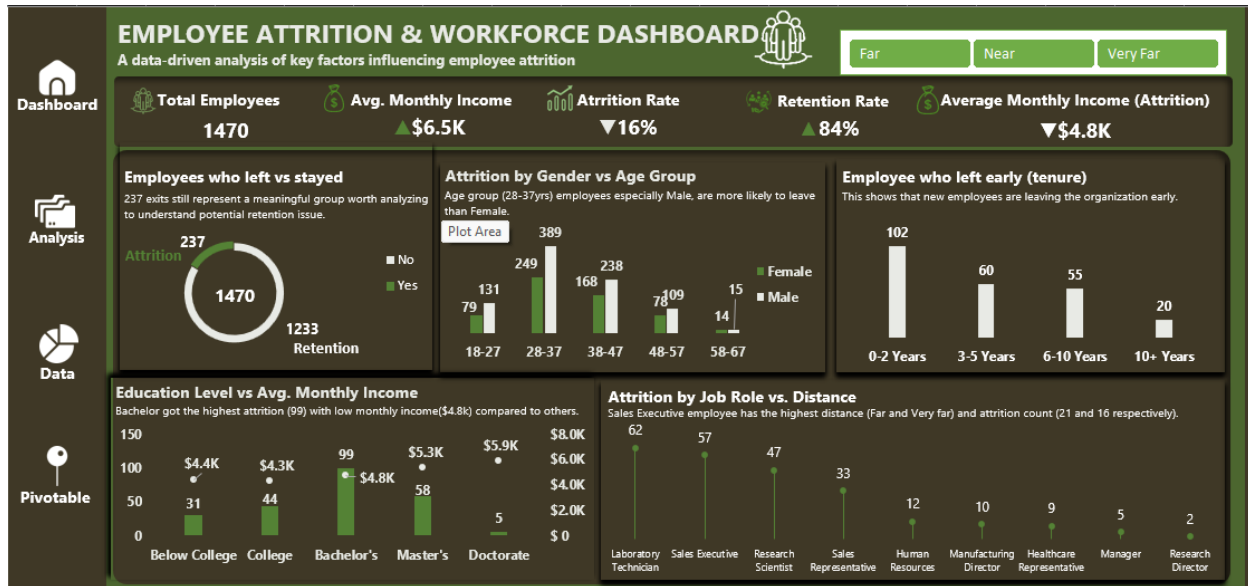
Employees reporting good work-life balance recorded the highest attrition at 127, suggesting that work-life balance alone is insufficient to retain employees.

Implication: Organizations need to address compensation, growth and recognition simultaneously alongside work-life balance initiatives.

Visual Insight & Dashboard

An interactive Excel dashboard was published comprising **2 pages** with dynamic slicers.

Page	Question Addressed	Visuals
Page 1 — Dashboard	What are the overall attrition metrics, gender, tenure and education patterns?	KPI Cards, Donut Chart, Bar Charts, Lollipop Chart
Page 2 — Analysis	What are the satisfaction, department, promotion, performance, overtime and work-life balance patterns?	Donut Charts, Bar Charts, Lollipop Chart



RECOMMENDATION

Based on the analysis, the following recommendations are proposed for organizations seeking to reduce employee attrition:

1. Review Compensation Structures

The income gap between attrited (\$4.8K) and retained (\$6.5K) employees is significant. Organizations should conduct regular salary benchmarking and ensure compensation is competitive especially for Bachelor degree holders and single employees.

2. Strengthen Onboarding and Early Engagement

With 102 employees leaving within their first 2 years, organizations must invest in structured onboarding programs, mentorship and early career support to improve retention during the critical early tenure period.

3. Address Overtime and Burnout

Overtime workers account for 53.59% of attrition. Organizations should implement workload management policies and monitor overtime patterns to prevent burnout and dissatisfaction.

4. Support Recently Promoted Employees

With 159 recently promoted employees leaving, organizations should provide clear role expectations, adequate support and transition plans for newly promoted staff.

5. Retain High Performing Employees

Excellent rated employees leaving at 84% is a significant talent drain. Organizations should introduce recognition programs, growth opportunities and performance-based incentives to retain top contributors.

CONCLUSION

This report analysed the HR Employee Attrition dataset of 1,470 employees using SQL and Excel to uncover the key drivers of employee turnover. With a 16% attrition rate, the analysis revealed that compensation gaps, early tenure challenges, overtime burnout, promotion dissatisfaction and low job satisfaction are the primary factors contributing to attrition. The Research & Development department and Sales Executive role were identified as the most affected areas. Through data-driven findings and actionable recommendations, this report provides organizations with a clear roadmap to reduce attrition, improve retention and build a more engaged workforce.